

ME302:2024-24 II

COURSE PROJECT REPORT

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1 Problem Description

We are asked to find the maximum scaling factor that can be used for a closed loop turbomachinery testing facility.

For testing purposes, the test-section and the upstream diffuser are redesigned and fabricated. The diffuser has an inlet at station 3 and an outlet at station 4, with the following given dimensions:

- Pipe diameter: $D = 0.6$ m
- Available diffuser length: $L = 1.5$ m

The facility must operate within these constraints:

- Maximum allowable pressure in any part of the facility: 250 kPa (structural requirement).
- $p_{05} \geq p_{01}$ to maintain constant inlet conditions and closed-loop flow.

The prototype length is $l_{\text{prototype}} = 0.2$ m, and for experimental feasibility, the model must maintain similarity conditions by matching the Mach number and Reynolds number. The target operating conditions are:

- Mach number: $M = 0.55$
- Reynolds number (based on length): $Re = 3.0 \times 10^6$

The geometric scaling factor is defined as:

$$\text{Scale} = \frac{l_{\text{model}}}{l_{\text{prototype}}} \quad (1)$$

where the objective is to determine the maximum possible scale ($0 < \text{Scale} \leq 1$) that meets the facility constraints and company requirements.

Additional given data for analysis includes:

- Test section inlet diameter: $D_4 = 2 \times l_{\text{model}}$
- Compressor operation data provided in files (`.txt`)
- Mass flow rate equation:

$$\dot{m} = \dot{m}_{\text{ref}} \left(\frac{p_{01}}{101.325 \text{ kPa}} \right) \sqrt{\frac{293 \text{ K}}{T_{01}}} \quad (2)$$

- Inlet stagnation temperature: $T_{01} = 293 \text{ K}$
- Air properties: $R = 287 \text{ J/kg.K}$, $\gamma = 1.4$
- Viscosity for Reynolds number calculation: $\mu_4 = 1.83 \times 10^{-5} \text{ Pa.s}$
- Pressure losses:

$$\Delta p_{023} = p_{02} - p_{03} = 1.5\% \times p_{02}$$

$$\Delta p_{045} = p_{04} - p_{05} = 5\% \times p_{04}$$

The analysis aims to determine the maximum scale that can be tested while meeting facility constraints, the corresponding inlet stagnation pressure p_{01} , and the compressor operating point, which will be located on the p_0 ratio vs. mass flow rate curve.

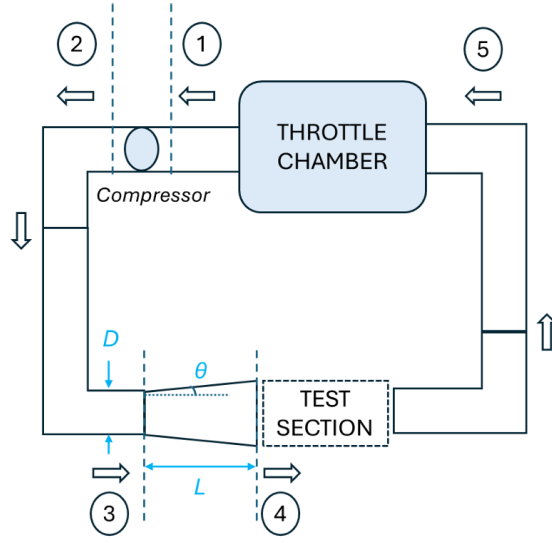


Figure 1: Facility Schematic

2 Methodology

2.1 Expression for Scaling Factor

- Expressing Reynold's number and the actual mass flow rate in terms of model parameters:

$$\frac{\rho_4 C_4 l_{\text{model}}}{\mu} = Re \quad \rho_4 \pi l_{\text{model}}^2 C_4 = \dot{m}$$

$$\implies Re \pi l_{\text{model}} \mu = \dot{m} = \dot{m}_{\text{ref}} \cdot \frac{p_{01}}{101.325} \sqrt{\frac{293}{T_{01}}}$$

- Incorporating prototype parameter $l_{\text{prototype}} = l_p$ (where $l_{\text{model}} = s \cdot l_p$):

$$K \cdot \frac{s}{\dot{m}_{\text{ref}}} = p_{01} \text{ where } K = 101.325 \times 10^3 \cdot Re \cdot \pi \cdot \mu \cdot l_p \cdot \sqrt{\frac{T_{01}}{293}}$$

$$\implies K \cdot p_0^r \cdot \frac{s}{\dot{m}_{\text{ref}}} = p_{02} \leq 250 \text{ kPa} \quad ; \quad p_0^r = \frac{p_{02}}{p_{01}} \quad , \quad T_0^r = \frac{T_{02}}{T_{01}}$$

- Incorporating pressure loss ($p_{03}/p_{02} = 0.985$) and neglecting viscous effects and losses across diffuser ($p_{03} = p_{04}$) :

$$0.985 \cdot K \cdot p_0^r \cdot \frac{s}{\dot{m}_{\text{ref}}} = p_{03} = p_{04}$$

- Incorporating Mach number (M) relations for static pressure and temperature values for:

1. Density (ρ_4):

$$\text{Define } Z = \left(1 + \frac{(\gamma-1)M^2}{2}\right) \longrightarrow T_4 = T_{04}/Z \quad p_{04} = p_4 \cdot Z^{\frac{\gamma}{\gamma-1}}$$

$$\frac{p_4}{RT_4} = 0.985 \cdot K \cdot p_0^r \cdot \frac{s}{\dot{m}_{\text{ref}} \cdot Z^{\frac{\gamma}{\gamma-1}} \cdot R \cdot \left(\frac{T_{04}}{Z}\right)} = 0.985 \cdot K \cdot p_0^r \cdot \frac{s}{\dot{m}_{\text{ref}} \cdot Z^{\frac{1}{\gamma-1}} \cdot R \cdot T_{04}} = \rho_4$$

2. Velocity (C_4):

$$C_4 = M \sqrt{\gamma R T_4} = \frac{M \sqrt{\gamma R T_{04}}}{\sqrt{Z}}$$

- Substituting ρ_4 , C_4 and K in Reynold's number expression:

$$\frac{\rho_4 C_4 l_{\text{model}}}{\mu} = Re = \frac{0.985 \times 101.325 \times 10^3 \times Re \times \pi \times \mu \times l_p^2 \times s^2 \times p_0^r \times M_4 \times \sqrt{\gamma}}{\dot{m}_{\text{ref}} \cdot Z^{\frac{\gamma+1}{2(\gamma-1)}} \cdot \mu \sqrt{293R}}$$

Assuming the facility to be adiabatic $\longrightarrow T_{02} = T_{03} = T_{04} = T_{05}$ and $T_{01} = 293$ K

- Rearranging gives us Scale (s):

$$s = \sqrt{\left(\frac{Z^{\frac{\gamma+1}{2(\gamma-1)}} \cdot \sqrt{293R}}{0.985 \times 101.325 \times 10^3 \times \pi \times l_p^2 \times M_4 \times \sqrt{\gamma}} \right) \times \left(\frac{\dot{m}_{\text{ref}} \sqrt{T_0^r}}{p_0^r} \right)}$$

2.2 Maximizing Scale Factor under constraints

The facility must operate within these constraints:

- Maximum allowable pressure in any part of the facility: 250 kPa (structural requirement).

We have the following relations between various pressures:

$$p_{02} = 1.0153 p_{03} = 1.0153 p_{04} = 1.0686 p_{05} = K \cdot p_0^r \cdot \frac{s}{\dot{m}_{\text{ref}}}$$

$$\implies p_{02} > p ; p \in \{p_{01}, p_{03}, p_{04}, p_{05}\} , K = 3.495 \times 10^6$$

$$\implies s < 7.1527 \times 10^{-2} \cdot \frac{\dot{m}_{\text{ref}}}{p_0^r}$$

- $p_{05} \geq p_{01}$ to maintain constant inlet conditions and closed-loop flow.

$$\implies p_{05} = \frac{p_{02}}{1.0686} \geq p_{01}$$

$$\implies \frac{p_{02}}{p_{01}} = p_0^r \geq 1.0686$$

2.3 Algorithm and Code

We have used `python` to code the calculation of s under the constraints mentioned above. Following are the steps to compute the scaling factor s :

1. The function s was separated into two components:

- A constant part C :

$$C^2 = \frac{Z^{\frac{\gamma+1}{2(\gamma-1)}} \cdot \sqrt{293R}}{0.985 \times 101.325 \times 10^3 \times \pi \times l_p^2 \times M_4 \times \sqrt{\gamma}}$$

- A variable part V , dependent on data from the compression map:

$$V^2 = \frac{\dot{m}_{\text{ref}} \sqrt{T_0^r}}{p_0^r}$$

2. For each entry in the compression map Excel data, we computed the expression $\frac{\dot{m}_{\text{ref}} \sqrt{T_0^r}}{p_0^r}$
3. The maximum value of V was selected. This represents the variable contribution to s .
4. The scaling factor s was calculated by multiplying the constant part with the maximum value of V :

$$s = C \times V_{\text{max}}$$

5. Finally, the calculated value of s was checked against the required constraints.

Here is the code:

```
1 import math
2
3 # --- Constants ---
4 g = 1.4          # Gamma
5 R = 287          # Gas constant [J/kg K]
6 lp = 0.2         # Characteristic length [m]
7 M = 0.55         # Mach number
8 Z = 1 + ((g - 1) * M * M / 2) # Intermediate variable for C
9 A1 = (g + 1) / (2 * (g - 1))  # Power for Z
10
11 # Step 1: Compute constant part C
12 C_squared = (math.pow(Z, A1) * math.sqrt(293 * R)) / (0.985 * 101.325 *
13               1000 * math.pi * lp**2 * M * math.sqrt(g))
14 C = math.sqrt(C_squared)
15
16 # --- Compression Map Data (dot{m}, T0_ratio, p0_ratio) ---
17 data = [
18     (10.55028375, 1.068519008, 1.210453952),
19     (10.54743366, 1.068558297, 1.210681035),
20     (10.5265642, 1.068532365, 1.21072001),
21     (10.54056709, 1.068554906, 1.210876663),
22     (10.53225492, 1.068622414, 1.212341445),
23     (10.52059919, 1.068581486, 1.212024862),
24     # Rest of data goes here...
```

```

24 ]
25
26 # Step 2: Compute V^2 for each entry
27 v_values = []
28 for m_dot_ref, T02_T01, P02_P01 in data:
29     V_squared = (m_dot_ref * math.sqrt(T02_T01)) / P02_P01
30     v_values.append((V_squared, m_dot_ref, T02_T01, P02_P01))
31
32 # Step 3: Select max V
33 V_squared_max, a_max, b_max, c_max = max(v_values, key=lambda x: x[0])
34 V_max = math.sqrt(V_squared_max)
35
36 # Step 4: Compute scale factor s
37 s = C * V_max
38
39 # Step 5: Compute final pressures
40 P02 = (101.325 * 54.9 * lp * math.pi * s * c_max) / a_max
41 P01 = P02 / c_max
42 P03 = 0.985 * P02
43 P04 = P03
44 P05 = 0.95 * P04
45
46 # --- Outputs ---
47 print("Scale (s) =", s)
48 print("P01 =", P01)
49 print("P02 =", P02)
50 print("P03 =", P03)
51 print("P04 =", P04)
52 print("P05 =", P05)

```

Listing 1: Python code to compute scale factor and pressures from compression map

3 Results and Discussion

The Python script outlined above was executed using compressor map data to compute the maximum allowable scaling factor s under the specified facility constraints. The computed values are summarized below:

- **Maximum Scaling Factor:** $s = 0.6179$
- **Inlet Stagnation Pressure:** $p_{01} = 204.70$ kPa
- **Compressor Exit Stagnation Pressure:** $p_{02} = 247.78$ kPa
- **Post-Loss Pressure after Diffuser Inlet:** $p_{03} = 244.06$ kPa
- **Diffuser Exit Pressure (assumed equal to p_{03}):** $p_{04} = 244.06$ kPa
- **Final Stagnation Pressure Before Compressor Inlet:** $p_{05} = 231.86$ kPa

All computed pressures respect the structural constraint of $p < 250$ kPa throughout the system. Additionally, the condition $p_{05} \geq p_{01}$ is satisfied, ensuring the feasibility of closed-loop flow.

The resulting scale factor of approximately **0.618** indicates that the model can be scaled to about **61.8%** of the prototype’s physical size while maintaining similarity conditions for Mach number and Reynolds number. This value balances facility safety, performance requirements, and experimental fidelity.

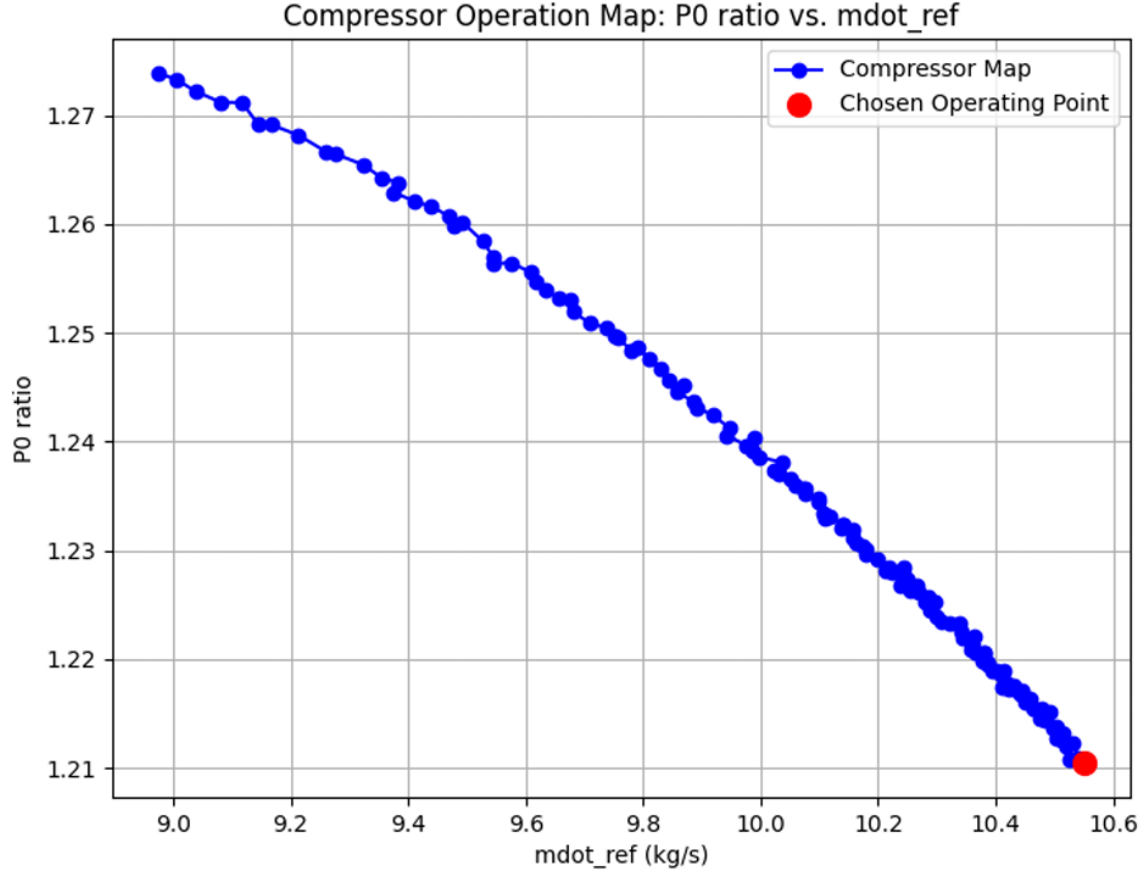


Figure 2: Compressor Map with design operating point

4 Future Work

The potential research and development opportunities are as follows:

Implementation and Validation of the jBAY Vortex Generator Model for Enhanced Flow Control in Turbomachinery Test Sections

This research offers a practical and efficient approach to improving flow control in turbomachinery systems through the implementation and validation of the jBAY vortex generator (VG) model. By simulating the effects of vortex generators—small surface-mounted devices that guide airflow—the model aims to reduce flow separation and enhance pressure recovery within diffusers and test sections. The simplified jBAY approach significantly lowers computational costs while maintaining accuracy, making it highly

suitable for integration into industrial design processes. Successful experimental validation could support the development of more fuel-efficient and high-performance systems in the aerospace and energy sectors. Additionally, it may attract the interest of software companies seeking to enhance CFD tools with robust and validated flow control models.

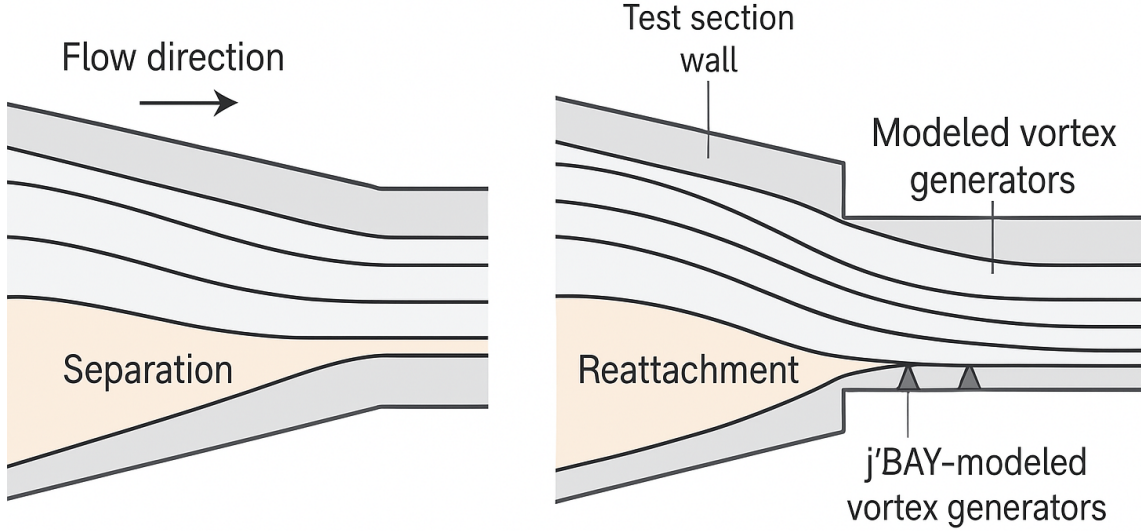


Figure 3: Illustration of the effect of jBAY-modeled vortex generators on diffuser flow behavior. The model enables efficient prediction and control of flow separation, contributing to improved aerodynamic performance in turbomachinery systems.

Pre-swirl for pressure recovery in conical diffusers

This research provides critical insight into how swirling inlet flow affects the performance of conical diffusers—components widely used in propulsion, turbomachinery, and fluid handling systems. The findings show that under certain conditions, introducing swirl can actually enhance pressure recovery and mitigate flow separation, especially in diffusers prone to stall under axial flow. This challenges the traditional view of swirl as purely detrimental and opens new avenues for design optimization. Such knowledge is particularly valuable for aerospace companies (e.g., GE Aerospace, Rolls-Royce) and industrial flow system manufacturers (e.g., Sulzer, Flowserve), where improved diffuser performance translates directly to higher system efficiency and lower operational costs. Additionally, these insights could guide the development of active flow control strategies and next-generation diffuser geometries for applications in compressors, pumps, and energy recovery systems.

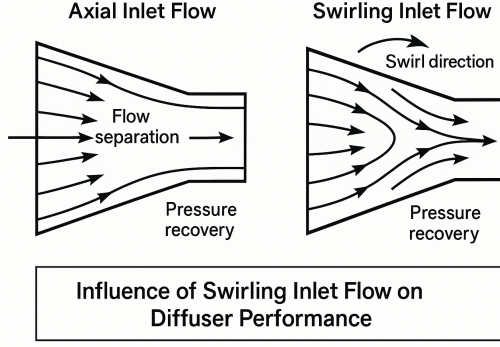


Figure 4: Effect of swirling inlet flow on flow behavior in a conical diffuser.

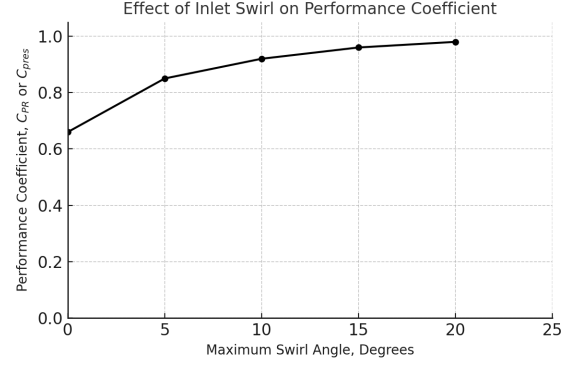


Figure 5: Effect of inlet swirl on performance coefficient for conical diffuser

The influence of varying area ratios (AR) and outlet-inlet configurations on pressure recovery and flow uniformity

This research highlights how 3D turning diffusers, when designed with carefully varied area ratios and inlet-outlet configurations, can significantly enhance pressure recovery and flow uniformity in low Reynolds number applications. The study recorded pressure recovery coefficients as high as 0.5, while maintaining flow uniformity with standard deviations in the range of $2.04\text{E-}03$ to $2.57\text{E-}03$. These promising results point toward the potential of geometrically optimized 3D diffusers to reduce energy losses and flow separation in compact duct systems.

Looking ahead, this work provides a strong experimental foundation for validating CFD models, enabling further exploration of diffuser shapes under different operating conditions. With precise flow visualizations and quantified performance metrics, this study opens opportunities for developing automated diffuser design tools, optimizing fluid pathways in HVAC, aerospace, turbomachinery, and industrial ventilation systems. Further investigation into a broader range of area ratios and curvature profiles could lead to the development of universal design guidelines for efficient diffuser integration in next-generation flow systems.

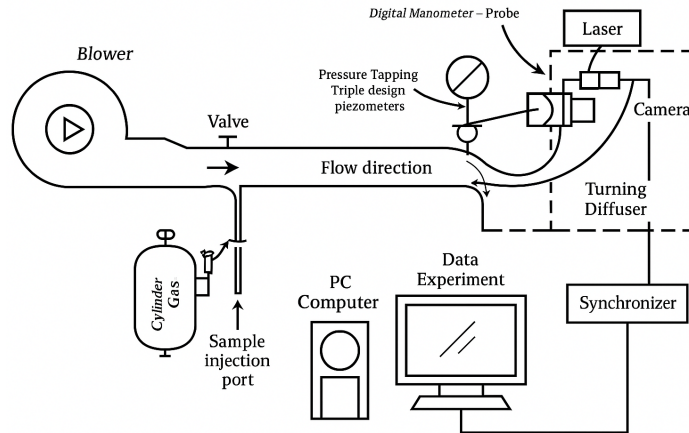


Figure 6: Example experimental setup for testing

5 Supplement

These are the file names and their links:

a) **Python File:** ME302_Python

b) **Latex zip file:** ME302_220844_220793_221034_Latex